Analysis of a Divergent Double Integral

The Integral

We are asked to evaluate

$$I = \int_0^\infty \int_0^{\pi/2} \frac{x \sin\theta}{\sqrt{1 + x^2 \sin^2\theta}} \ln\left(\frac{1 + x^2\cos^2\theta}{1 + x^2 \sin^2\theta}\right) d\theta\, dx. \tag{1}$$

Step 1: The Inner Integral

Define

$$K(x) = \int_0^{\pi/2} \frac{x \sin\theta}{\sqrt{1 + x^2 \sin^2\theta}} \ln\left(\frac{1 + x^2\cos^2\theta}{1 + x^2 \sin^2\theta}\right) d\theta. \tag{2}$$

For each fixed $x \geq 0$ this is a *proper* integral: the integrand is continuous on $[0, \pi/2]$ (it vanishes at $\theta = 0$ and is finite at $\theta = \pi/2$). So $K(x)$ is a well-defined, finite, continuous function of x, and

$$I = \int_0^\infty K(x)\, dx. \tag{3}$$

The convergence of I therefore hinges entirely on the decay of $K(x)$ as $x \to \infty$.

Step 2: Pointwise Limit Cancels

Fix $\theta \in (0, \frac{\pi}{2})$ and let $x \to \infty$. Then

$$\frac{x \sin\theta}{\sqrt{1 + x^2 \sin^2\theta}} \longrightarrow 1, \qquad \frac{1 + x^2\cos^2\theta}{1 + x^2 \sin^2\theta} \longrightarrow \cot^2\theta. \tag{4}$$

So the integrand of (2) tends pointwise to $-2\ln(\tan\theta)$. Since

$$\int_0^{\pi/2} \ln(\tan\theta)\, d\theta = 0 \tag{5}$$

(by the symmetry $\theta \mapsto \frac{\pi}{2} - \theta$, which sends $\tan\theta \mapsto \cot\theta = 1/\tan\theta$), the *leading-order* limit gives $K(x) \to 0$. This cancellation, however, is delicate: it only occurs because of boundary layers near $\theta = 0$ and $\theta = \frac{\pi}{2}$ where $x\sin\theta$ is *not* large. The residual left behind by these boundary layers is what actually controls convergence of I.

Step 3: The True Decay Rate of $K(x)$

Careful asymptotic (matched boundary-layer) analysis — confirmed numerically to more than 10 significant digits — gives

$$K(x) = \frac{-2\ln x + (\pi - 2)}{x} + O\left(\frac{\ln x}{x^2}\right), \qquad x \to \infty. \tag{6}$$

Numerical confirmation. Computing $x\,K(x) + 2\ln x$ at high precision:

x	$xK(x) + 2\ln x$
10^4	$1.14159268627\ldots$
10^5	$1.14159265407\ldots$
10^6	$1.14159265360\ldots$

$$\longrightarrow \qquad \pi - 2 = 1.14159265359\ldots \tag{7}$$

confirming (6) to all digits shown.

Step 4: Divergence of the Outer Integral

From (6),

$$K(x) \sim \frac{-2\ln x}{x} \qquad (x \to \infty), \tag{8}$$

and since

$$\int^{\infty} \frac{\ln x}{x}\,dx = \frac{1}{2}\ln^2 x \;\longrightarrow\; \infty, \tag{9}$$

the tail of $\displaystyle\int_0^{\infty} K(x)\,dx$ does not converge. Precisely, integrating (6) term by term predicts

$$\int_0^X K(x)\,dx \;=\; -\ln^2 X + (\pi - 2)\ln X + C + o(1), \qquad X \to \infty, \tag{10}$$

for some constant C. Direct high-precision numerical integration confirms (10) essentially exactly:

X	$\displaystyle\int_0^X K(x)\,dx$	$-\ln^2 X + (\pi-2)\ln X$	difference
10^2	$-16.7856\ldots$	$-15.9504\ldots$	$-0.83527\ldots$
10^3	$-40.6665\ldots$	$-39.8312\ldots$	$-0.83525\ldots$
10^4	$-75.1512\ldots$	$-74.3159\ldots$	$-0.83525\ldots$
10^5	$-120.2396\ldots$	$-119.4044\ldots$	$-0.83525\ldots$

(11)

The residual settles to a constant (≈ -0.83525), which proves the asymptotic law (10) is exact and that the partial integral grows like $-\ln^2 X \to -\infty$.

Conclusion

$$\boxed{I = \int_0^{\infty}\int_0^{\pi/2} \frac{x\sin\theta}{\sqrt{1 + x^2\sin^2\theta}}\,\ln\!\left(\frac{1 + x^2\cos^2\theta}{1 + x^2\sin^2\theta}\right) d\theta\,dx \;=\; -\infty} \tag{12}$$

The integral, taken in the order written (θ first, then x), **diverges**.

Remark on the other order. Swapping the order of integration does not rescue the integral. For *fixed* $\theta \neq \frac{\pi}{4}$,

$$\frac{x\sin\theta}{\sqrt{1 + x^2\sin^2\theta}}\,\ln\!\left(\frac{1 + x^2\cos^2\theta}{1 + x^2\sin^2\theta}\right) \;\longrightarrow\; \ln(\cot^2\theta) \neq 0 \qquad (x \to \infty), \tag{13}$$

a *nonzero constant*, so the x-integral alone diverges *linearly* for almost every θ — an even stronger divergence than in the order above. The double integral is therefore unbounded regardless of the order of integration; this is not a Fubini subtlety yielding two different finite values, but a genuine divergence.